



Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: <http://www.clinicalnutritionespen.com>

Original article

In vitro digestion study comparing a predigested glycerolysis product versus long-chain polyunsaturated fatty acid-rich oils (LCPUFA) as a strategy for administering LCPUFA to preterm neonates

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ARTICLE INFO

Article history:

Received 1 December 2023

Accepted 13 September 2024

Keywords:

Arachidonic acid

Docosahexaenoic acid

Bioaccessibility

Enzymatic glycerolysis

In vitro digestion

LCPUFA

ABSTRACT

Background & aims: Maintaining an adequate supply of arachidonic acid (ARA) and docosahexaenoic acid (DHA) is essential for optimal growth of preterm infants. This study aims to evaluate and compare the digestibility and bioaccessibility of ARA and DHA oils compared to their predigested product through an *in vitro* digestion model.

Methods: An *in vitro* gastrointestinal digestion model was used in two stages: gastric digestion and intestinal digestion. Samples of two polyunsaturated rich oils (ARA and DHA oils) and their predigested product (2:1, ARA: DHA) produced by enzymatic glycerolysis have been digested for 120 min. The final digestion product obtained was composed of three phases: an upper oily phase (OP) containing the undigested species, an intermediate micellar phase (MP) containing digested and bioaccessible lipids, and a precipitate phase (PP) with insoluble compounds. The reaction was monitored by taking aliquots and their subsequent lipid extraction and analysis.

Results: Poorer digestibility for ARA and DHA oils was observed based on the percentage of the oily phase (26.7% and 20%, respectively) found compared to the glycerolysis product (GP) oily phase (13.9%). The highest micellar phase was found in the GP (approx. 83%). On the other hand, the monoglyceride (MAG) content was lower in the digestion product (DP) from ARA and DHA oils, 4.3% and 9.2%, respectively, compared to the MAG observed in the DP of GP (15%).

Conclusion: Considering the percentage of oily phase, micellar phase, and the MAG content, it can be concluded that the GP is more digestible and ARA and DHA are more bioaccessible than in its precursor oils.

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1. Introduction

In recent years, a great interest has been brought about arachidonic acid (ARA) and docosahexaenoic acid (DHA), especially for newborns because these fatty acids has a crucial activity for visual

and cognitive development during the embryogenesis [1,2]. Particularly, in preterm newborns, there is a reduced blood level of ARA and DHA that has been associated with neonatal morbidities [3]. The fetus receives DHA and ARA through placental transfer in an ARA: DHA ratio ca. 2:1 [4], therefore, adding DHA and ARA to preterm infant formula in this right ratio may improve the infant's ability to respond to immunological challenges as well as visual and cognitive development in a comparable way to breastfed infants [5]. Based on their relevance, it has been recommended supplementation of these two fatty acids in infant formula by different institutions (i.e. Food and Agricultural Organization (FAO)/World Health Organization (WHO) [6].

Bioaccessibility and bioavailability are crucial aspects when discussing the benefits of ARA and DHA. In this sense, the chemical

Abbreviations: ARA, Arachidonic acid; DHA, Docosahexaenoic acid; LCPUFA, long-chain polyunsaturated fatty acids; FFA, free fatty acid; MAG, monoacylglyceride; DAG, diacylglyceride; TAG, triacylglyceride; GP, glycerolysis product; DP, digestion product; OP, oil phase; MP, micellar phase; PP, precipitate phase.

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<https://doi.org/10.1016/j.clnesp.2024.09.011>

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form in which these fatty acids are administered could be an appropriate strategy. It has been described that triglycerides (TAGs) containing long-chain polyunsaturated fatty acids (LCPUFA) are more resistant, at least *in vitro*, to pancreatic lipase than other edible oils [7]. The oral bioavailability of polyunsaturated fatty acids strongly depends on the chemical form utilized [8]. It has been suggested that diacylglyceride (DAG) and monoacylglyceride (MAG) molecules, may facilitate micellar formation and therefore improve long-chain polyunsaturated fatty acid (LCPFA) absorption into the enterocyte [9,10]. The supramolecular organization of lipids has been also related to oral bioaccessibility [11–13]. Furthermore, for oral bioaccessibility, adequate pancreatic lipase levels should be also considered. Pancreatic failure is related to several diseases such as cystic fibrosis (CF), Shwachman Syndrome, and Johanson-Blizzard Syndrome produced enzymatic deficiencies such as pancreatic triglyceride lipase, whose deficiency has also been observed in newborns and infants [14].

For all these reasons utilization of predigested forms of LCPUFA oils is a critical prerequisite when trying to improve its bioaccessibility. These predigested forms can be hydrolysis products or glycerolysis products (GP) from different LCPUFA oils. This is the only way to circumvent LCPUFA oral bioaccessibility limitations previously described. Pre-digested forms can bypass lower pancreatic lipase activity towards marine oils and improve emulsification during digestion in conditions with impaired pancreatic activity. Regarding these two types of predigested forms of edible oils, hydrolysis products resulting from a reaction in the presence of water, contain free fatty acids to a bigger extent, which is detrimental because of their hydrolytic rancidity. GP is produced by an enzymatic glycerolysis, which can be defined as an alcoholysis using glycerol as an acyl-acceptor catalyzed by different lipases. This procedure has been extensively studied for the production of MAG and/or DAG from animal and vegetable oils [15] and the final products are extensively used in the food industry, mainly as emulsifiers. In addition, an important aspect to highlight is that in recent years, they have been reviewed and accepted by the European Food Safety Authority (EFSA) as food additives [16].

Finally, regarding premature newborns, it is essential to know the physiology and the specific conditions of digestion they undergo when optimizing *in vitro* digestion models [17,18]. As they are fed through enteral or parenteral feeding, they do not have oral digestion, so it is crucial to focus on their gastrointestinal tract [19]. They have also higher gastric pH, bile salt concentrations, different intestinal barriers and gastrointestinal motility compared to a healthy newborn. In particular, lipid digestion is essential for their growth and development because is the main source of energy for them [18]. The poor digestion of triglycerides and their accumulation in the intestines have been shown to lead to inflammatory diseases such as necrotizing enterocolitis [20]. Therefore, the administration of pre-digested lipids is important to alleviate or reduce the incomplete digestion of triglycerides and the adverse effects it entails [21].

The aim of this work is to compare the digestibility of ARA oil, DHA oil and the enzymatic glycerolysis product produced from the mixture of ARA and DHA oils in a 2:1 (w/w) ratio by means of an *in vitro* digestion model to establish which product is more digestible and could be a good candidate for the development of LCPUFA-rich infant formulas with good bioaccessibility.

2. Materials and methods

2.1. Materials

Arachidonic fatty acid-rich oil (ARA oil) from animal sources was supplied by Penta Manufacturing Company (Caldwell, NJ, United

States) and docosahexaenoic fatty acid-rich oil (DHA oil) sourced from the marine microalgae *Schizochytrium* sp. was supplied by Progress Biotech (Capelle aan den IJssel, Netherlands), both oils were stored in a modified atmosphere of nitrogen at -20°C . Reagents used for *in vitro* digestion, trizma, pepsin, maleic acid, bile salts (BS), and cholesterol were purchased from Sigma–Aldrich (St. Louis, MO, USA). Phosphatidylcholine from egg yolk (PC) was supplied by Lipoid (Ludwigshafen, Germany), food-grade phospholipase A2 (PLA2) from *Streptomyces violaceoruber* (103 U mg^{-1}) was supplied by Nagase Chemtex Corporation, Fukuchiyama Factory (Kyoto, Japan). Pancreatin from porcine pancreas was purchased from MP Biomedicals, LLC (Irvine, USA). Hydrochloric acid, sodium chloride, and calcium chloride were from Panreac (Barcelona, Spain). Rabbit gastric lipase was supplied by Lipolytech (Marseille, France) and stored in freezer storage. Hexane (HEX), methyl-tertbutyl ether (MTBE), petroleum ether (PE), and chloroform were supplied from Macron (Avantor Performance Materials, Center Valley, USA), and formic acid (98% purity) from Panreac (Barcelona, Spain). All these solvents were of HPLC and GC grade.

Pure standards of ARA oil and DHA oil (MAG and DAG) were obtained by Solid Phase Extraction (SPE) using silica gel from Supelco–Merck Group (Darmstadt, Germany) from their glycerolysis products. The step-elution was carried out according to the method described by Ingalls [22].

2.2. Methods

2.2.1. Glycerolysis reaction

Glycerolysis product (GP) was obtained by lipase-catalyzed glycerolysis carried out according to the method of Torres et al. [22] with slight modifications. Briefly, 10 g of ARA oil and DHA oil mixture with a ratio 2:1, were added to a 120 mL flask containing 1 g of glycerol. The mixture was thermostated at 40°C with continuous agitation (200 rpm). The reaction was started with the addition of Novozyme 435 in an enzyme-to-oil ratio of 1:10 (w/w). Then, the flask containing the reaction mixture was incubated in an orbital shaker at 200 rpm and 40°C (IKA KS 4000 ic Control, Staufen, Germany). The reaction was left for 48 h, and the reaction product was recovered by centrifugation at 40°C and 10,000 rpm for 15 min.

2.2.2. Characterization of fatty acid profile by gas chromatography

The fatty acid (FA) profile of the samples (ARA oil, DHA oil, and GP) under study was determined by gas chromatography (GC) with a flame-ionization detector (GC-FID), prior methylation of the esterified fatty acid residues present in the samples. For this, 50 mg of samples were dissolved with 1 mL of n-hexane. Then, 1 mL of 0.5 NaOH in methanol was added for sample methanolysis, followed by stirring for 1 min, heating at 100°C for 10 min, and further methylation by adding 1 mL of 14% BF_3 in methanol (Supelco, Pasadena, USA), stirring for 1 min and incubation at 100°C for 10 min. The resulting mixture of fatty acid methyl esters (FAMES) was extracted with n-hexane (1 mL) after stirring for 2 min at room temperature. The decanted hexane phase was then allowed to stand for 2 h with anhydrous sodium sulfate (Panreac) for humidity elimination. Finally, hexane was evaporated until dryness at 40°C under a nitrogen stream by using a Stuart Block Heater SBH200D/3 (Staffordshire, U.K.) to obtain a residue of FAMES which was then redissolved in n-hexane at a final concentration of 12 mg mL^{-1} .

Chromatographic analysis of FAMES was performed on an Agilent Technologies gas chromatograph (6850 N Network GC System) equipped with a flame ionization detector (FID). A sample of $1\text{ }\mu\text{L}$ was injected in the split/splitless mode (split ratio 10:1) in a capillary column HP-88 ((88%-cyanopropyl) aryl-polysiloxane, $30\text{ m} \times 0.25\text{ mm} \times 0.20\text{ }\mu\text{m}$ thickness (Agilent Technologies Inc.,

Santa Clara, CA, USA). Helium was used as carrier gas at a flow rate of 0.9 mL min⁻¹. Injector and detector temperatures were 220 and 250 °C, respectively. The temperature program was as follows: starting at 50 °C and then heating to 180 °C at 20 °C min⁻¹; followed by heating from 180 to 220 °C at 15 °C min⁻¹. The final temperature (220 °C) was held for 10 min. Identification of the free fatty acids was based on PUFA no. 3 standard obtained from Sigma–Aldrich (Steinheim, Germany). These determinations were carried out in duplicate.

2.2.3. Lipid profile by gas chromatography

The lipid profile of the ARA oil, DHA oil, and GP was determined by GC analyses in an Agilent gas Chromatograph 7820A (Agilent Technologies, Santa Clara, CA, USA) with on-column injection coupled to an FID detector, according to the method described by Torres et al. [23]. The used column was an HP-5MS capillary column, 5% phenyl methyl silicone (length 7 m, internal diameter 0.25 mm, and thickness 0.25 µm). The injection volume was 0.1 µL. The temperature of the injector and detector was 50 and 340 °C, respectively. The program of temperatures started at 60 °C, increasing at 42 °C min⁻¹ until 250 °C. This temperature was maintained for 20 min and then increased up to 340 °C at 25 °C min⁻¹, which was maintained for 35 min. The quantification was performed by the external standard method using pure standards. For this, calibration curves were made with each of the standards. The data is expressed as a percentage of TAG, DAG, MAG, FFA, and Cholesterol (%species) concerning the weight of the residue (w.r.) obtained after lipid extraction, according to the following equation:

$$\% \text{ of species} = (W_{\text{comp}} / W_{\text{w.r.}}) \times 100 \quad [1]$$

Where W_{comp} is the weight of each type of lipid and $W_{\text{w.r.}}$ is the weight of the residue obtained after lipid extraction.

2.2.4. In vitro gastrointestinal digestion model

An *in vitro* gastrointestinal digestion model was carried out based on the study from Chabni et al. [7] with some modifications, one first stage of gastric digestion and a second stage of intestinal digestion.

2.2.4.1. Gastric digestion. Fat samples of ARA oil, DHA oil and GP (1.86 g) were pre-emulsified with 7.44 mL of gastric phase simulation fluid (SFG) in a digester jacketed with temperature control at 37 °C and constant stirring at 900 rpm for 10 min, using a mechanical stirrer (Metrohm, Herisau, Switzerland), after sonication with ultrasound probe (Vibra cell, VC 130 (Sonics, Newtown, Connecticut, United States) for 1 min at 70%. Gastric digestion began by adding 1.16 mL of a solution previously prepared, consisting of 400 mg of rabbit gastric lipase (activity >15 U mg⁻¹ of protein), 24 mg of porcine pepsin (EC 3.4.23.1) in 2.5 mL of distilled water, which was kept under stirring at 900 rpm for 10 min. Gastric digestion was carried out for 60 min at 37 °C and 900 rpm.

To study the evolution of lipid products during the hydrolytic process that takes place in gastric digestion, aliquots of 400 µL were taken at minutes 10, 20, 30, 40, and 60. At minute 60 a sample of 2.925 mL was also taken, to start with 1 g of fat in intestinal digestion.

2.2.4.2. Intestinal digestion. After gastric digestion, 26 mL of Trizma-maleate buffer (0.1 M, pH 7.5) was added to the stirred mixture. To simulate biliary secretion, a solution was prepared consisting of 500 mg of bile salts, 200 mg of egg yolk phosphatidylcholine, 40 mg of cholesterol (Sigma–Aldrich Chemie GmbH, Steinheim, Germany), 1 mL of a 325 mM CaCl₂ solution, 3 mL of a 3.25 mM NaCl solution (Panreac Química S.A.U, Barcelona, Spain)

and 20 mL of 0.1 M Trizma-maleate buffer, pH 7.5. This mixture was homogenized with ultraturrax T18 basic IKA (Staufen, Germany) for 2 min at 3500 rpm. Afterward, the simulation bile secretion was added to the resulting emulsion after gastric digestion, and all of it was homogenized again with ultraturrax for 2 more minutes at 3500 rpm. The digestion intestine began by adding a solution of fresh pig pancreatin extract prepared from 100 mg of pancreatin in 4 mL of Trizma-maleate buffer 0.1M pH 7.5, stirred at 900 rpm for 10 min, and centrifuged at 1600 G, at 5 °C for 15 min. The supernatant was added to the reaction medium together with 2.5 mL of phospholipase A2 food grade. The intestinal phase was carried out for 60 min at 37 °C and 900 rpm.

To study the evolution of lipid products during the hydrolytic process that takes place in intestinal digestion, aliquots of 1 mL were taken at minutes 2, 5, 10, 20, 30, 45, and 60 being one of the final digestion products (DP). In addition, an aliquot was also taken at minute 0 of the intestinal digestion, i.e., before adding the pancreatin solution.

2.2.5. Phase separation after in vitro digestion

After 60 min of intestinal digestion, the digestion products were centrifuged at 4500 rpm and 37 °C for 45 min (Thermo Scientific Sorvall 6000 LYNX) according to the method described by Martin et al. [24]. After centrifugation, three phases were obtained: an upper oily phase (OP), formed by the undigested lipid fraction; an intermediate aqueous fraction, called micellar phase (MP), which contains the lipid fraction digested in form of micellar and vesicular structures, and a precipitate phase (PP), which consists of a lower phase containing insoluble compounds at 37 °C. These three phases were separated from each other for the subsequent extraction of the lipids present in each of them.

2.2.6. Lipid extraction

The total lipids contained in the three phases obtained (OP, MP, and PP), the aliquots taken to study the evolution of digestion, and the final digestion products (DP), were extracted from the samples in three sequential steps using solvent mixtures with increasing polarity in a solvent/sample ratio of 3:1 (v/v) and centrifuging for 10 min at 14,500 rpm each time [25], except PP, which is extracted with a solvent/sample ratio of 6:1 (v/v). The solvents used were: n-hexane:methyl-tert-butyl-ether (MTBE) (50:50, v:v); MTBE: Petroleum ether (PE) (50:50, v:v); and PE: ethanol (1:0.6, v:v). After each extraction, the supernatant was collected in an empty vial and possible impurities were allowed to decant. Later, the supernatant was transferred to another previously weighed vial and evaporated under a stream of nitrogen using a Stuart Block Heater SBH200D/3 (Staffordshire, U.K.) to constant the weight of residue (w.r.). Finally, the samples were diluted with acidified chloroform (0.1 % formic acid) to a final concentration of 6 mg mL⁻¹ before injection in the GC system.

2.2.7. Lipid identification and quantification by gas chromatography

Lipid characterization of lipid residue of the aliquots taken to study the evolution of digestion, OP, MP, PP, and DP, was carried out by gas chromatography according to the method described in section 2.2.3. After composition characterization, the distribution of each lipid species was determined by combining the relative weight of each phase and composition according to the following equation:

$$\% \text{ of A} = \text{eight percent of oil phase} \times (\text{percentage of A in oil phase} / 100) + \text{weight percentage of the micellar phase} \times (\text{percentage of A in the micellar phase} / 100) + \text{weight percentage of the precipitated phase} \times (\text{percentage of A in the precipitated phase} / 100) [2]$$

Where A refers to any of the types of lipids analyzed in this study.

2.2.8. Statistical analysis

Data analysis was performed using Excel (Microsoft Office) and all statistical evaluations were performed using Origin (version 9.0 for Windows; OriginLab Corporation, Northampton Northampton, USA). Experiments were carried out by duplicate, and the data were expressed as mean ± standard deviation. The statistical significance of the differences between the groups was measured by one-way analysis of variance (ANOVA) and post-hoc Tukey HSD test. Statistical significance was defined at the level of $p < 0.05$.

3. Results

3.1. Fatty acid profile of arachidonic fatty acid-rich oil (ARA oil), docosahexaenoic fatty acid-rich oil (DHA oil), and glycerolysis product (GP)

Table 1 shows the fatty acid composition of the oils under study. As it can be observed similar level of LCPUFA was observed in three products (66–68%). Microalgae oil (MO) used as a source of DHA contains 54.9% of this acid. In addition, the presence of other omega-3 LPUFA like docosapentaenoic acid (DPA) stands out at 10%, and a slightly lower level of monounsaturated fatty acids was attained. Regarding ARA-rich oil, it accounts for 53.8% of the total oil. Other types of omega-6 such as linoleic acid (6.2%) and γ -linolenic acid (4.9%) are also found in smaller quantities. The fatty acid profile of GP verifies the 2:1 ratio of the mixture of two oils, ARA and DHA.

3.2. Lipid composition of arachidonic fatty acid-rich oil (ARA oil), docosahexaenoic fatty acid-rich oil (DHA oil), and glycerolysis product (GP)

The lipid profile of the ARA and DHA oils and the GP was determined and the results are shown in Table 2. It can be seen that in the case of the ARA and DHA oils most of the lipid species are TAGs, present in 89.8% and 95.1%, respectively. A self-emulsifying system was created by mixing ARA and DHA oils in a ratio of 2:1, respectively. This lipid delivery system was produced via enzymatic glycerolysis, to be utilized as a pre-digested product in this study.

Table 1
Fatty acid profile of Arachidonic fatty acid-rich oil (ARA oil), Docosahexaenoic fatty acid-rich oil (DHA oil), and glycerolysis product (GP) of a mixture 2:1 (w/w) of the two LPUFA oils. Significant differences ($p < 0.05$) in the same raw are indicated with different letters a, b, or c. Values are means ± SD.

Fatty acid (g/100g)	ARA oil	DHA oil	GP
Myristic acid (C14:0)	—	0.67 ± 0.03 ^a	0.48 ± 0.00 ^b
Myristoleic (C14:1)	—	0.64 ± 0.04 ^a	0.24 ± 0.00 ^b
Palmitic acid (C16:0)	7.08 ± 0.08 ^a	17.61 ± 1.85 ^b	10.4 ± 0.03 ^c
Palmitoleic acid (C16:1)	0.51 ± 0.03 ^a	0.43 ± 0.04 ^a	0.47 ± 0.03 ^a
Stearic acid (C18:0)	11.1 ± 0.04 ^a	1.71 ± 0.1 ^b	7.6 ± 0.08 ^c
Oleic acid (C18:1)	8.26 ± 0.04 ^a	9.36 ± 1.09 ^a	8.1 ± 0.08 ^a
Linoleic acid (C18:2)	6.23 ± 0.04 ^a	1.33 ± 0.21 ^b	4.4 ± 0.03 ^c
γ -Linolenic acid (C18:3)	4.92 ± 0.20 ^a	0.33 ± 0.04 ^b	2.9 ± 0.06 ^c
α -Linolenic acid (C18:3)	0.61 ± 0.01 ^a	0.22 ± 0.02 ^b	0.56 ± 0.06 ^a
Arachidonic acid (20:4)	53.8 ± 0.32 ^a	—	36.4 ± 0.07 ^b
Eicosapentaenoic acid (C20:5)	—	0.74 ± 0.04 ^a	0.47 ± 0.01 ^b
Behenic acid (C22:0)	4.70 ± 0.00 ^a	1.46 ± 0.15 ^b	3.30 ± 0.02 ^c
Docosapentaenoic acid (C22:5)	—	10.0 ± 0.33 ^a	4.04 ± 0.09 ^b
Docosahexaenoic acid (C22:6)	—	54.9 ± 0.51 ^a	18.7 ± 0.03 ^b
Others	2.88 ± 0.07 ^a	1.93 ± 0.15 ^b	1.53 ± 0.10 ^c
Total saturated FA	19.9 ± 0.03 ^a	21.9 ± 2.00 ^a	19.0 ± 0.13 ^a
Total monounsaturated FA	13.8 ± 0.51 ^a	10.4 ± 1.00 ^b	12.7 ± 0.07 ^a
Total polyunsaturated FA	66.3 ± 0.4 ^a	67.7 ± 1.00 ^a	68.3 ± 0.06 ^a

Glycerolysis product (GP) is a mixture of acylglycerols comprised of ca. 20% of MAGs, ca. 50% of DAGs and 30% of TAGs.

3.3. In vitro gastrointestinal digestion of ARA oil

Figure 1A depicts the time course of *in vitro* gastric and intestinal digestion of ARA oil. After gastric digestion, the hydrolysis of ARA oil TAGs in the gastric phase is lower than 10%. When intestinal digestion begins, the first stage of a rapid rate of hydrolysis (corresponding to the first 5 min of the intestinal stage, and a slower hydrolysis rate is observed in the rest of the intestinal stage. Approximately, 63% of fatty acids are released at the end of the *in vitro* digestion, corresponding to 80% of the hydrolysis of TAGs.

3.3.1. Digestion product

After the intestinal phase, the digestion product (DP) was extracted and analyzed by gas chromatography. This product was comprised of ca. 62% free fatty acids, 4% MAGs, 5% cholesterol, 7% DAGs, and 20% TAGs (see Fig. 2A). It should be noted that there is an abundant amount of undigested TAG and DAG (27%) and a very low amount of MAGs (4%). A suitable digestion product is considered to contain at least 20% w/w MAGs and ≤10% oil phase (TAGs and DAGs) [7].

The DP was also centrifuged, and three phases were obtained (Fig. 2B), namely the oily phase (OP), micellar phase (MP), and precipitate phase (PP). In this case, ca. 70% (w/w) of the digestion product was found in MP, which is considered the bioaccessible phase. Remarkably, more than 26% of DP was OP, which is a very abundant non-digestible fraction. Finally, a very small PP (3.8%), which contains insoluble lipid materials from the digestion product, was also attained. Regarding the phase's lipid composition (Fig. 2C), OP was formed mainly by TAGs (56.3%) and DAGs (19.9%) that were not digested. MP consisted mainly of FFAs (78.8%) with a small percentage of MAGs (4.9%) and TAGs (7.7%).

3.3.2. Distribution of each lipid compound among the different phases

It is important to know how each lipid class distributes in each phase (Fig. 2A) to know which lipid compounds are more bio-accessible. Therefore, it is necessary to correlate the percentage of each species in the DP (Fig. 2C), the proportion of each phase obtained after centrifugation (Fig. 2B), and the percentage of each lipid class in each phase. As an example, regarding FFA, the composition of OP indicates that we have ca. 18.9% of FFA in this phase and this phase represents 26.7% of the total digestion product. Hence, to know the FFA content in OP regarding total FFA, we combine these two numbers, and we obtain that, 5 parts from a total 62 parts of FFA in digestion product, are in OP. If we express this value in percentage, it corresponds to ca. 8% of total FFA in OP. In addition, by summing up the content of all lipids in each phase, one obtains the proportion of each phase obtained after centrifugation.

Table 2
Lipid composition (%) of Arachidonic fatty acid rich oil (ARA oil), Docosahexaenoic fatty acid rich oil (DHA oil) and glycerolysis product (GP). Significant differences ($p < 0.05$) in the same raw are indicated with different letters a, b, or c. Values are means ± SD.

Lipids (g/100g)	ARA oil	DHA oil	GP
Tryglicerides (TAG)	89.82 ± 0.08 ^b	95.09 ± 0.03 ^c	28.26 ± 0.0 ^a
Diglycerides (DAG)	4.30 ± 0.05 ^a	4.27 ± 0.04 ^a	50.64 ± 0.0 ^b
Monoglycerides (MAG)	0.33 ± 0.08 ^a	0.26 ± 1.85 ^a	18.21 ± 0.03 ^b
Free fatty acids (FFA)	4.68 ± 0.03 ^c	0.24 ± 0.04 ^a	2.27 ± 0.03 ^b
Cholesterol (CHOL)	0.87 ± 0.04 ^b	—	0.63 ± 0.08 ^a

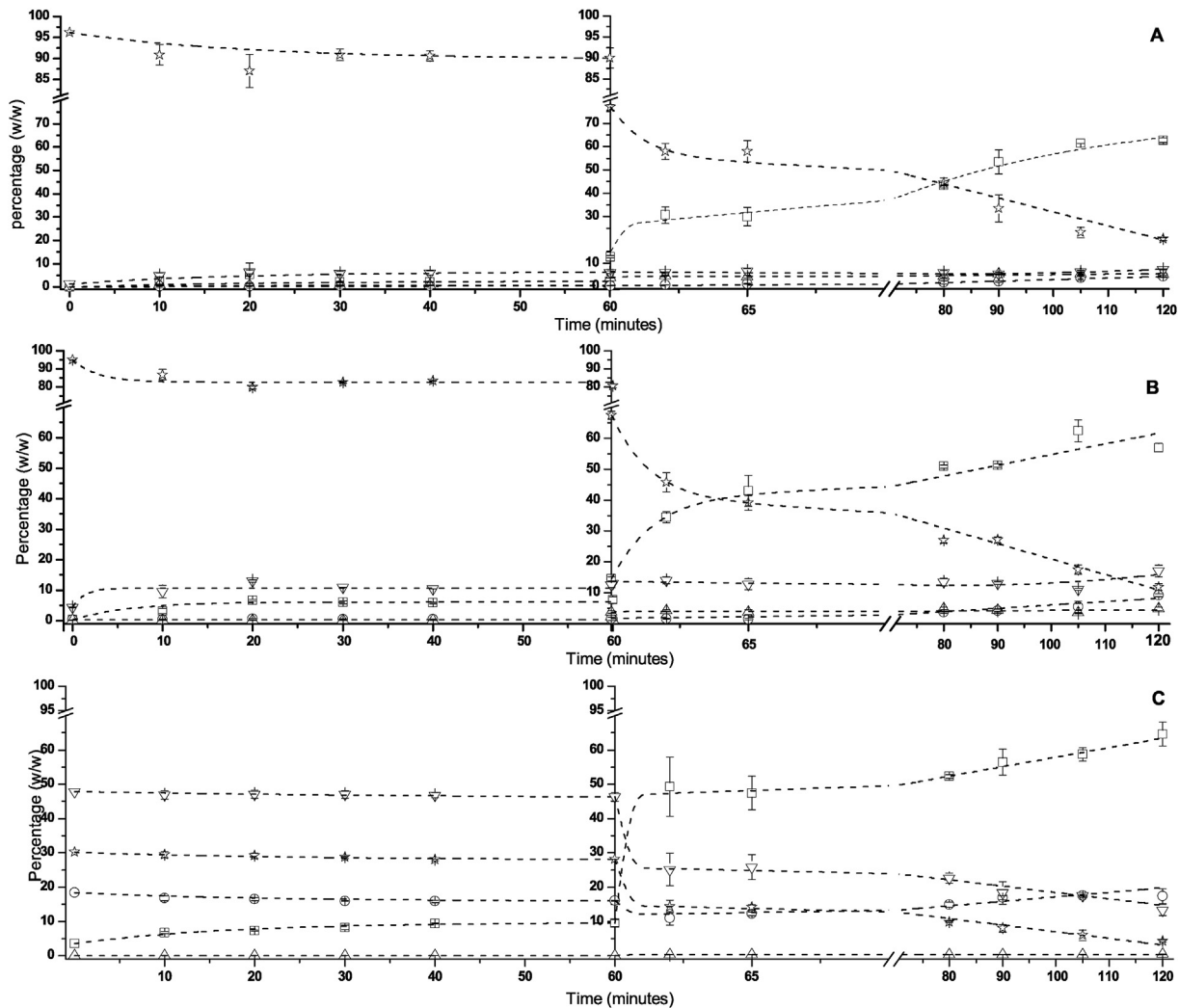


Fig. 1. Time course of *in vitro* gastric and intestinal digestion of Arachidonic acid (ARA) rich oil (1A), Docosahexaenoic acid (DHA) rich oil (1B), and glycerolysis product from Arachidonic acid (ARA) and Docosahexaenoic acid (DHA) rich oils mixture in a ratio 2 to 1 (w/w). □ FFA: free fatty acid, ○ MAG: monoacylglyceride, △ Cholesterol, ▽ DAG: diacylglyceride, ☆ TAG: triacylglyceride.

If we want to know the distribution of TAGs among three phases, we proceed similarly. Therefore, if we have 20.3% of TAGs in the DP and 15 parts of these 20.3 parts are in OP, 72.8% of total TAGs are in OP. It can be concluded that FFA, MAGs, and cholesterol are found mainly in MP, and DAGs and TAGs are located mainly in OP.

3.4. *In vitro* gastrointestinal digestion of DHA oil

As can be observed in Fig. 1B, the hydrolysis of TAGs in the gastric phase is 10%, similar to the gastric phase of ARA oil. In the intestinal phase, again, two different rates of hydrolysis can be observed. A first stage of a rapid rate of hydrolysis (corresponding to the first 5 min of the intestinal stage), and a slower hydrolysis rate observed in the rest of the intestinal stage. Approximately, 60% of fatty acids are released at the end of the *in vitro* digestion, corresponding to 85% of hydrolysis of TAGs.

3.4.1. Digestion product

Final DP of DHA oil was comprised of ca. 57% free fatty acids, 9% MAG, ca. 5% cholesterol, 17% DAG, and ca. 12% TAG (Fig. 3A). Except for cholesterol, these values are significantly different ($p < 0.05$) from those obtained for the ARA oil digestion product. There is an

abundant quantity of TAGs and DAGs in DP that indicates poor digestibility. Surprisingly, there is a low amount of MAGs, although is higher than that observed in the digestion product of ARA oil ($p = 3.4 \cdot 10^{-5}$).

Regarding the three phases obtained after DP centrifugation (Fig. 3B), similarly to ARA oil, 70% (w/w) of DP was found in MP, which is considered the bioaccessible phase. It should be noted that 20% of the digestion product was OP which is an abundant non-digestible fraction. Surprisingly, an abundant PP (10%), comprised of insoluble lipid materials in the digestion product, was attained. Other than cholesterol ($p = 0.006$) and MAGs ($p = 0.01$), the composition of this phase was similar to MP, but it was comprised of lipid classes insoluble in MP or solid at 37 °C. Regarding the composition of OP, it was formed mainly by DAGs (41%) and TAGs (28.4%) that were not digested. MP consisted mainly of FFAs (64%) and MAGs (12%) and also contained DAGs (11.3%) and TAGs (8.1) (see Fig. 3C).

3.4.2. Distribution of each lipid compound among the different phases

Figure 3A also depicts lipid class distribution among OP, MP, and PP after *in vitro* digestion of DHA oil. As expected, most of the FFA

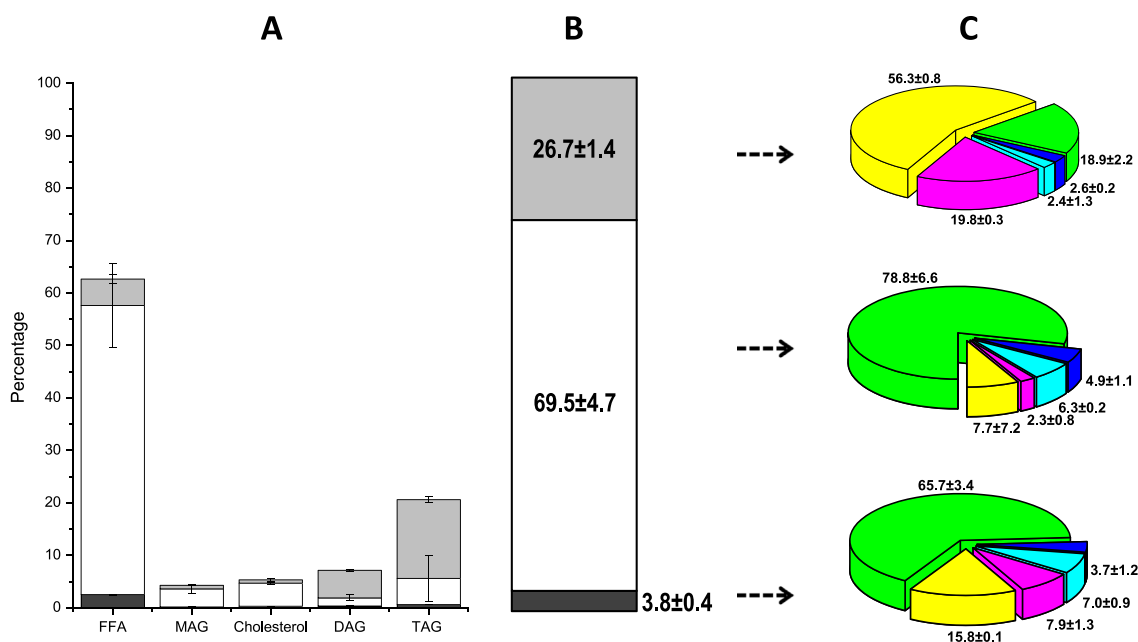


Fig. 2. Lipid composition of OP, MP and PP (A), distribution of digestion product among OP, MP and PP (B), composition of digestion product and classes distribution among OP, MP and PP (C) after *in vitro* digestion of Arachidonic acid (ARA) rich oil. FFA: free fatty acid, MAG: monoacylglyceride, Cholesterol, DAG: diacylglyceride, TAG: triacylglyceride, OP: oil phase, MP: micellar phase, PP: precipitate phase.

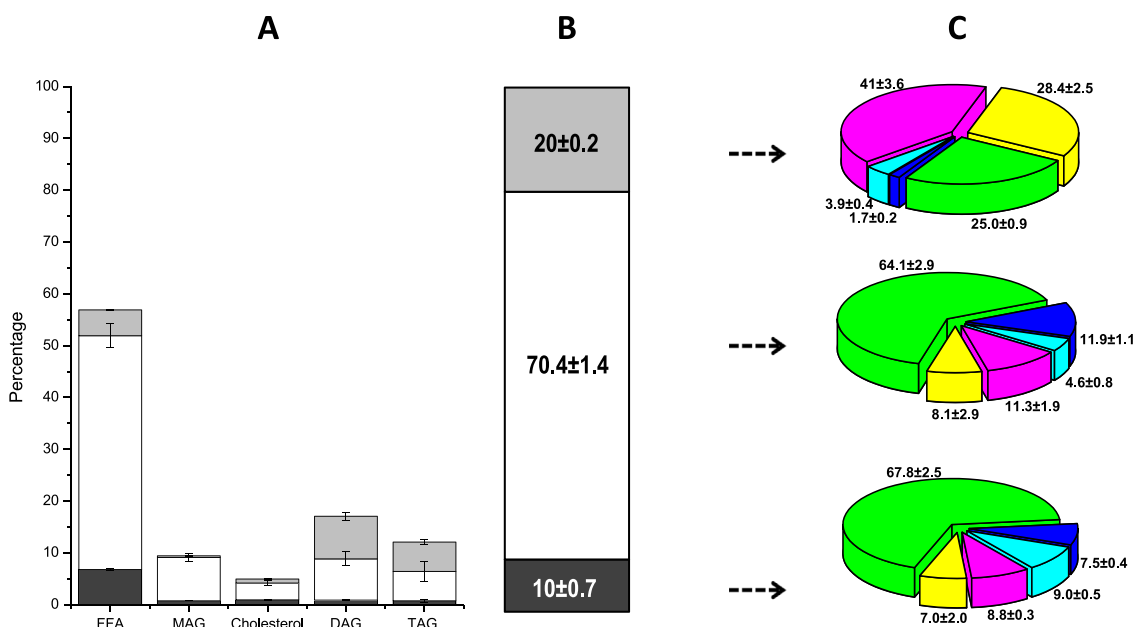


Fig. 3. Lipid composition of OP, MP and PP (A), distribution of digestion product among OP, MP and PP (B), composition of digestion product and lipid classes distribution among OP, MP and PP (C) after *in vitro* digestion of Docosahexaenoic acid (DHA) rich oil. FFA: free fatty acid, MAG: monoacylglyceride, Cholesterol, DAG: diacylglyceride, TAG: triacylglyceride, OP: oil phase, MP: micellar phase, PP: precipitate phase.

and MAG are located in MP, ca. 80% and 90%, respectively. Particularly noteworthy are the DAGs and TAGs that are distributed between the MP (46 and 47 %, respectively) and the OP (48 and 47 %, respectively).

3.5. *In vitro* gastrointestinal digestion of a GP from a mixture of ARA and DHA oils

Similarly to the previous *in vitro* digestions investigated, the time course of GP was studied. The results are depicted in Fig. 1C. It

can be observed that the gastric phase produced a very small extent of hydrolysis releasing ca. 10% of FFA that comes from the different acylglycerols contained in the glycerolysis product. On the contrary, along the intestinal phase, ca. 60% of FFA was released.

3.5.1. Digestion product

The digestion product contained less than 5 % of TAGs and around 15% of DAGs. Besides, the content of MAGs was ca. 15% (Fig. 4A). These are indicators of a much more adequate digestibility of glycerolysis products in comparison with their precursor oils.

After centrifugation, 83% of DP was found in MP, which is considered the bioaccessible phase (Fig. 4B). This micellar phase was higher than that observed with the precursor oils. It should be noted that ca. 14% of DP was OP which is an acceptable non-digestible fraction. Very low PP (3.8%), comprised of insoluble lipid materials in the digestion product, was also attained. The composition of this phase was similar to MP but comprised of lipid classes insoluble in MP or solid at 37 °C. Regarding the composition of OP (Fig. 4C), it was formed mainly by DAGs (65.8%) and TAGs (21%) that were not digested. MP consisted mainly of FFA (67.5%) and MAGs (17.2%) and also contained small proportions of DAGs (8.2%) and TAGs (1.8%).

3.5.2. Distribution of each lipid compound among the different phases

Figure 4A shows lipid class distribution among OP, MP, and PP after *in vitro* digestion of GP. It should be noted that MP contained more than 90% of both FFA and MAGs. It is also worth noting that 64 and 56% of TAGs and DAGs are in OP, respectively. Similarly to DHA oil, an abundant proportion of total DAGs and TAGs are incorporated in MP, which also indicates better emulsification properties of MP compared to that observed in MP from ARA oil.

4. Discussion

4.1. Fatty acid profile and composition of ARA oil, DHA oil, and glycerolysis product (GP)

LCPUFAs composition of the oils under study is similar to those obtained in the literature for the same species of microalgae oil (51 and 9.4% of DHA and DPA, respectively) [25]. Also, for ARA oil, similar contents of arachidonic acid and linoleic acid have been reported in the literature, present in 46.2% and 5.0%, respectively, in ARA-rich oils of fungal origin [26]. These profiles make both oils (ARA and DHA oils) exceptionally significant natural sources of

omega-6 and omega-3 LCPUFAs. The fatty acid profile of GP confirms the 2:1 ratio of the mixture of the two oils.

Regarding the lipid composition of the ARA and DHA oils, it is mainly composed of TAGs. The lipid composition of ARA oil has not been found in the bibliography, and that of DHA oil is similar to that obtained by the same authors in previous studies [7,27]. As for the self-emulsifying system created by mixing ARA and DHA oils in a ratio of 2:1, it has a similar acylglycerols composition to that obtained by other authors [15,27]. Similar values of DAGs and MAGs content of GP with LCPUFA have been found in the literature [28]. Also, it should be noted that GP presented only 2.2% of FFA, which indicates the stability of this product in terms of oxidation of free fatty acids.

4.2. *In vitro* gastrointestinal digestion of ARA oil, DHA oil, and glycerolysis product (GP)

Hydrolysis extent is very low through the gastric phase of the three oils under study. In adults, gastric lipase hydrolysis releases between 10 and 25% of fatty acids [29]. It has been also reported that gastric lipase has a preference towards short and medium-chain fatty acids discriminating against long-chain fatty acids [30]. In addition, gastric lipase is active in a wide range of pH (from 2 to 8) and continues hydrolyzing TAGs at the intestinal phase [31]. This phenomenon can be observed in Fig. 1 in the gap between the end of the gastric phase and the beginning of the intestinal phase that starts when pancreatic lipase is added. The disappearance of TAGs in this period could be caused by gastric lipase that is still active when pH is elevated to reach pH 7, necessary for the optimum activity of pancreatic lipase. Also in the intestinal phase, two different rates of hydrolysis can be observed for the oils under study.

Regarding ARA oil, after the intestinal phase, the digestion product (DP) was extracted. The abundant amount of TAG and DAG indicates poor digestibility. Surprisingly, there is a very low quantity of MAGs that could indicate that MAGs could be hydrolyzed as

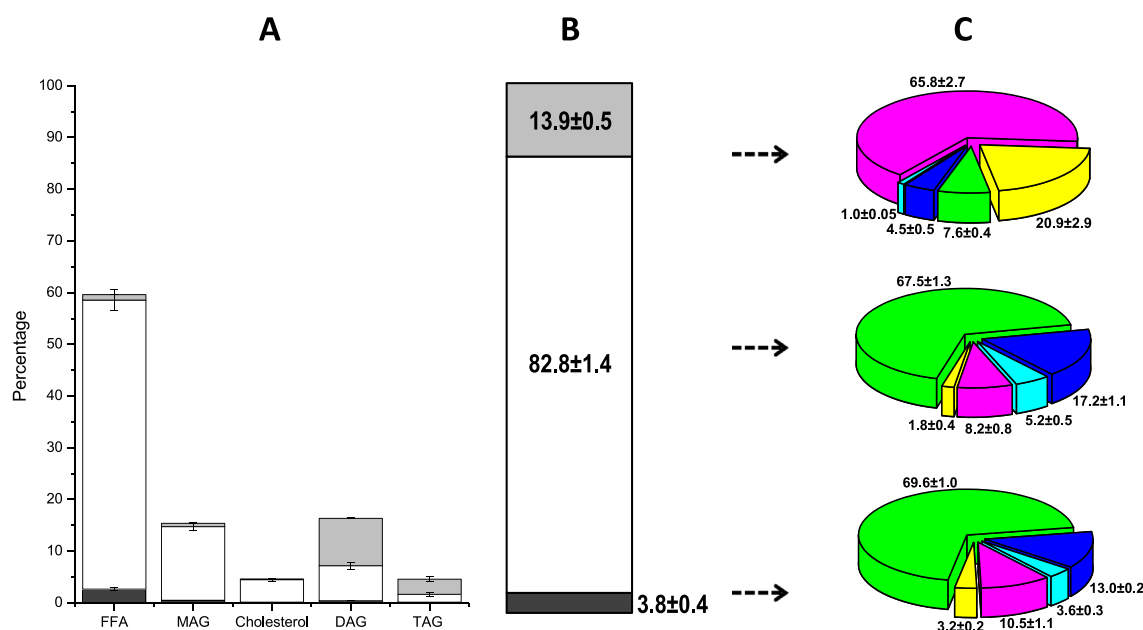


Fig. 4. Lipid composition of OP, MP and PP (A), distribution of digestion product among OP, MP and PP (B), composition of digestion product and lipid classes distribution among OP, MP and PP (C) after *in vitro* digestion of glycerolysis product obtained from Arachidonic acid (ARA) and Docosahexaenoic acid (DHA) rich oils mixture in a ratio of 2 to 1 (w/w). FFA: free fatty acid, MAG: monoacylglyceride, Cholesterol, DAG: diacylglyceride, TAG: triacylglyceride, OP: oil phase, MP: micellar phase, PP: precipitate phase.

soon as it is produced releasing free fatty acids and glycerol. This hydrolysis could be carried out by carboxy ester hydrolase [32]. There are few studies regarding hydrolysis of lower glycerides by digestive lipases [24,33,34]. This low amount of MAGs in the digestion product could be indicative of inadequate digestibility of this oil since FFAs and MAGs should be the major lipid forms after digestion. As a premise, we consider that an adequate digestion product should contain around 20% w/w of MAGs and $\leq 10\%$ of oily phase [35].

Regarding the gastric phase of DHA oil, the hydrolysis extent of TAGs is similar to that of ARA oil and in agreement with the results obtained by Aarak et al. [36] in the *in vitro* digestion of salmon oil rich in EPA and DHA. Moreover, in the study carried out by Na & Lee [37], in which they did a comparative study of the *in vitro* digestion of microalgae oil, fish oil, and soybean oil, it was found that after *in vitro* digestion of the three oils, microalgae oil had the poorest digestibility, due to a lower release of FFA after 2 h of digestion. These authors explain these results in the microalgae oil because of the higher polyunsaturation of FFA located in sn-1,3 and the double bond close to the carboxyl group that it presents, which inhibits the TAG hydrolysis activity of lipase, resulting in lower digestion rates of microalgal oil. Regarding the three phases obtained after centrifugation of DP, the amount of OP is reduced compared to that obtained in a previous study in which gastric lipase was not utilized in which OP was 27% [8]. This indicates the additional extent of hydrolysis carried out by gastric lipase and could simulate a more realistic scenario of digestion.

Regarding the digestibility of the GP, it is digested to a bigger extent than the precursor oils and contains 83% of MP (the bio-accessible phase). Moreover, a much lower amount of OP and a higher level of MAG, which are indicators of better digestibility and bioaccessibility of this product compared to its precursor oils. These results suggest that the presence of an emulsifier in the medium promotes lipid digestion in agreement with Guo, Bellissimo [38]. Emulsifying lipids boost lipid dispersion at the start of digestion, which may improve both TAGs and DAGs hydrolysis and the creation of micellar structures, which allow DP to be more easily absorbed. This has been also reported in previous works of *in vitro* digestion of the GP of shark liver oil [25]. Finally, GP can be considered for several clinical applications since it is a pre-digested product that improves bioaccessibility, as a self-emulsifying system that allows its incorporation in formula milk, or as a nutritional supplement for enteric administration for premature newborns [21], since it is a system that releases bioactive lipids by incorporating ARA and DHA with proven biological activity [6].

5. Conclusion

It could be concluded that based on monoacylglyceride (MAG) content and percentage of oily phase (OP), ARA and DHA oils have inadequate digestibility. Considering an adequate digestion product, it should contain around 20% w/w of MAGs and no more than 10% of OP. These two criteria are not fulfilled in ARA oil and DHA oil digestion products. Instead, this study demonstrates that utilization of a predigested product such as the glycerolysis product (GP) utilized in the present study, can be a much more effective way to administrate ARA and DHA to prevent inadequate pancreatic lipase activity, inadequate hydrolysis extent or emulsification problems, limitations that can be found using the precursor oils, i.e., ARA and DHA oils.

Funding statement

This study has been funded by Fundación Ramón Areces (CIVP20A6607) and Ministerio de Ciencia e Innovación (project

number PID2020-119084RB-C21). Blanca Pardo de Donlebún thanks to Comunidad de Madrid Programa Yo Investigo 2022 supported by the Spanish Recovery, Transformation and Resilience Plan-NextGeneration EU. Celia Bañares thanks the Ministerio de Economía y Competitividad and the European Social Fund for a pre-doctoral FPI grant (BES-2017-080,853). Assamae Chabni also thanks Fundación de la Universidad Autónoma de Madrid for her pre-doctoral Research contract.

Author contributions

All authors conducted the experimentation equally, Assamae Chabni, Blanca Pardo de Donlebún, and Celia Bañares conducted the data processing and Carlos Torres wrote the main manuscript text. All authors reviewed and edited the manuscript.

Data availability statement

The data obtained for the publication of this article are available upon reasonable request to the corresponding author.

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